

MLFF Architecture Revision 54

PERF-P5 TRAIN2/EVAL2 persistence and reuse hardening

mdstats development

2026-08-15

Revision 54

Revision 54 closes the CPU/control-plane portion of **PERF-P5** without changing scientific authority. The release is mdstats 0.20.187a0; dependency-graph schema 36 is synchronized with this manual revision.

TRAIN2/STOR2 persistence

TRAIN2 live/EMA-state hashing and STOR2 evaluation-capsule hashing retain the existing canonical metadata and contiguous tensor-byte sequence. The implementation now feeds those bytes to SHA-256 through bounded memoryview chunks instead of creating an additional full-size bytes object through `numpy().tobytes()` solely for hashing.

For a state sequence $T = (t_1, \dots, t_n)$, the authority remains

$$H = \text{SHA256}(\text{bar.v.double}[D_i \parallel Q_i \parallel B_i]),$$

with unchanged schema marker S , dtype metadata D_i , shape metadata Q_i , and canonical tensor bytes B_i .

TRAIN2 additionally writes execution-only `train2_persistence.jsonl` records that separate clone, tensor-hash, raw-checkpoint-hash, continuation-write, summary-write, and total persistence costs. Continuation state is unchanged.

EVAL2 compatible-model reload

A candidate-only, unaccelerated, uncompiled MACE shell may opt into strict compatible-state reload. The path fails closed unless model class, complete state-key set, tensor shapes and tensor dtypes match exactly; state application is strict. Source-foundation-bound, CuEq/OEq, and compiled providers do not use this path. Fresh model construction remains the default.

The supplied MH-1 CPU/e3nn comparison is prediction-exact but the reload path is 6.49% slower than fresh construction on this host. It is therefore not promoted as a CPU default. Any accelerator benefit must be established in FINAL-GPU1.

Measured CPU result

On a 256 MB contiguous FP32 state, two fresh-process samples per path show:

Authority	Pre-P5 median	PERF-P5 median	Change	Extra RSS before	Extra RSS after
TRAIN2 state digest	265.02 ms	142.97 ms	46.05% lower	245.00 MiB	0.75 MiB
STOR2 cap- sule digest	248.19 ms	146.87 ms	40.82% lower	244.94 MiB	0.81 MiB

Old and new digests are byte-identical in both cases.

Dataset-format boundary

No HDF5/LMDB conversion is introduced. Those formats remain MACE dataset-storage mechanisms; they are not represented as authenticated reusable AtomicData graph caches. Existing DATA8 fixed-file reuse and DATA6/EVAL2 graph/prediction caches remain authoritative.

Roadmap

All remaining accelerator-dependent evidence is held for the one-shot **FINAL-GPU1** package. CUEQ-DEP1, the CuEq phase gates, E3NN-BASELINE, accelerator-side PERF-P5 comparison, and PERF-CERT1 therefore remain final-release obligations rather than intermediate development interruptions.

References

- Python Software Foundation, *Built-in Types - Memory Views*, Python 3 documentation: <https://docs.python.org/3/library/stdtypes.html#memory-views>.
- PyTorch, *Saving and Loading Models* and `torch.nn.Module.load_state_dict`: https://docs.pytorch.org/tutorials/beginner/saving_loading_models.html; https://docs.pytorch.org/docs/stable/generated/torch.nn.Module.html#torch.nn.Module.load_state_dict.
- MACE documentation, *Heterogeneous Data Training and Large Dataset Pre-processing*: https://mace-docs.readthedocs.io/en/latest/guide/heterogeneous_data.html; <https://mace-docs.readthedocs.io/en/latest/guide/multipreprocessing.html>.